



Binomial Theorem

If x and y are variables and n is a positive integer, then

$$(x+y)^n = \binom{n}{0} x^0 y^n + \binom{n}{1} x^1 y^{n-1} + \binom{n}{2} x^2 y^{n-2} + \dots + \binom{n}{n} x^n y^0 = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Note: (i) nC_k or $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ $0 \leq k \leq n$.

(ii) $nC_0 = nC_n = 1$ and $nC_1 = n$.

(iii) $nC_k = nC_{n-k}$.

1. Determine the Coefficient of $x^5 y^2$ in the expansion of $(x+y)^7$.

Sol: we know that $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$.

Therefore the Coefficient of $x^k y^{n-k}$ in the expansion of $(x+y)^n$ is $\binom{n}{k}$. Hence the Coefficient of $x^5 y^2$ in the expansion of $(x+y)^7$ is $\binom{7}{5} = 21$.

2. Determine the Coefficient of $x^2 y^2$ in the expansion of $(x+y)^4$ (Ans: 6)

3. Determine the Coefficient of $x^9 y^3$ in the expansion of $(x+y)^{12}$ (Ans: 220)

4. Determine the Coefficient of $a^5 b^2$ in the expansion of $(2a-3b)^7$.

Sol: we know that $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$.

Let $x=2a$ and $y=-3b$.

$$\therefore (2a-3b)^n = \sum_{k=0}^n \binom{n}{k} (2a)^k (-3b)^{n-k}$$

$$= \sum_{k=0}^n \binom{n}{k} 2^k (-3)^{n-k} a^k b^{n-k}$$

Therefore the Coefficient of $a^k b^{n-k}$ in the expansion of $(2a-3b)^n$ is $\binom{n}{k} 2^k (-3)^{n-k}$. Hence the Coefficient of $a^5 b^2$ in the expansion of $(2a-3b)^7$ is $\binom{7}{5} (2)^5 (-3)^2 = 6048$.

5. Determine the Coefficient of $x^9 y^3$ in the expansions of
 (i) $(x+2y)^{12}$ (ii) $(2x-3y)^{12}$ (Ans (i) 1760 (ii) -3041280)



Multinomial Theorem

For positive integers n, t , the Coefficient of $x_1^{n_1} x_2^{n_2} x_3^{n_3} \dots x_t^{n_t}$ in the expansion of $(x_1 + x_2 + x_3 + \dots + x_t)^n$ is $\frac{n!}{n_1! n_2! n_3! \dots n_t!}$,

Where each n_i is an integer with $0 \leq n_i \leq n$, for all $1 \leq i \leq t$ and $n_1 + n_2 + \dots + n_t = n$.

1. Determine the Coefficient of $x^2 y^2 z^3$, xyz^5 and $x^3 z^4$ in the expansion of $(x+y+z)^7$.

Sol: The Coefficient of $x^2 y^2 z^3$ in the expansion of $(x+y+z)^7 = \frac{7!}{2!2!3!} = 210$.

The Coefficient of xyz^5 in the expansion of $(x+y+z)^7 = \frac{7!}{1!1!5!} = 42$

The Coefficient of $x^3 z^4$ in the expansion of $(x+y+z)^7 = \frac{7!}{3!0!4!} = 35$.

2. Determine the Coefficient of xyz^2 in $(x+y+z)^4$ (Ans: 12)

3. Determine the Coefficient of xyz^2 in $(w+x+y+z)^4$ (Ans: 12)

4. Determine the Coefficient of xyz^2 in the expansion of $(ax+by+cz)^4$

Sol: By multinomial theorem the Coefficient of $x_1^{n_1} x_2^{n_2} x_3^{n_3}$ in the expansion of $(x_1 + x_2 + x_3)^n$ is $\frac{n!}{n_1! n_2! n_3!}$, provided $n_1 + n_2 + n_3 = n$.

Let $x_1 = ax$, $x_2 = by$, $x_3 = cz$.

$\therefore$ the Coefficient of $x_1^{n_1} x_2^{n_2} x_3^{n_3}$ in the expansion of $(ax+by+cz)^n$ is $\frac{n!}{n_1! n_2! n_3!} \cdot (a)^{n_1} (b)^{n_2} (c)^{n_3}$. Hence the Coefficient of xyz^2

in the expansion of $(ax+by+cz)^4$ is $\frac{4!}{1!1!2!} (a)^1 (b)^1 (c)^2$
 $= 12abc^2$

5. Determine the Coefficient of (i) xyz^2 in $(2x-y-z)^4$ (Ans: -24)
 (ii) xyz^2 in $(x-2y+3z)^4$ (Ans: -216)
 (iii) $w^3x^2yz^2$ in $(2w-x+3y-2z)^8$ (Ans: 161280)

6. Determine the Coefficient of $a^2b^3c^2d^5$ in the expansion of $(a+2b-3c+2d+5)^{16}$.

Sol: we know that the Coefficient of $x_1^{n_1}x_2^{n_2}x_3^{n_3}x_4^{n_4}x_5^{n_5}$ in the expansion of $(x_1+x_2+x_3+x_4+x_5)^n$ is $\frac{n!}{n_1!n_2!n_3!n_4!n_5!}$, provided,

$$n_1+n_2+n_3+n_4+n_5=n.$$

$$\text{Let } x_1=a, x_2=2b, x_3=-3c, x_4=2d, x_5=5$$

$$\text{and } n_1=2, n_2=3, n_3=2, n_4=5, n_5=n-(n_1+n_2+n_3+n_4)$$

$$= 16-(2+3+2+5) = 4.$$

$$\text{Hence the Coefficient of } a^2b^3c^2d^5 \text{ in the expansion of } (a+2b-3c+2d+5)^{16}$$

$$= \frac{16!}{2!3!2!5!4!} (1)^2(2)^3(-3)^2(2)^5(5)^4$$

$$= 4.35891456 \times 10^{14}$$

7. Determine the Coefficient of $w^2x^2y^2z^2$ in the expansion of
- (i) $(w+x+y+z+1)^{10}$ (Ans: 113400)
- (ii) $(2w-x+3y+z-2)^{12}$ (Ans: 718502400)
- (iii) $(v+w-2x+y+5z+3)^{12}$ (Ans: 1.010394×10^{10})

* For each integer $n > 0$, P.T (i) $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$.
(ii) $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0$.

pf: (i) From the binomial theorem,

$$\sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = (x+y)^n \quad \text{--- (1)}$$

put $x=y=1$ in eq(1) we get

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

$$\text{i.e. } \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n.$$

That is, Sum of all the Coefficient in the expansion of $(x+y)^n$ is 2^n .

(ii) Put $x=-1, y=1$ in eq(1), we get

$$\sum_{k=0}^n \binom{n}{k} (-1)^k = 0$$

$$\text{i.e. } \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + (-1)^n \binom{n}{n} = 0.$$

1. Determine the Sum of all the Coefficients in the expansion of $(x+y)^3$

Sol: we know that the Sum of all the Coefficients in $(x+y)^n$ is 2^n .

Hence the Sum of all the Coefficients in $(x+y)^3$ is $2^3 = 8$.

2. Determine the Sum of all the Coefficients in the expansions of
(i) $(x+y)^{10}$ (ii) $(x+y)^7$.
(Ans: (i) 2^{10} (ii) 2^7)

* Note: The Sum of all the Coefficients in the expansion of $(a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_tx_t)^n$ is $(a_1 + a_2 + a_3 + \dots + a_t)^n$.

3. Determine the Sum of all the Coefficients in the expansions of
(i) $(x+y+z)^{10}$ (ii) $(w+x+y+z)^5$ (iii) $(2s-3t+5u+6v+11w+3z+2y)^{10}$
(Ans: (i) 3^{10} (ii) 4^5 (iii) 4^{10})

1. For any positive integer n determine (i) $\sum_{i=0}^n \frac{1}{i!(n-i)!}$ (ii) $\sum_{i=0}^n \frac{(-1)^i}{i!(n-i)!}$

$$\text{Sol: (i) } \sum_{i=0}^n \frac{1}{i!(n-i)!} = \sum_{i=0}^n \frac{n!}{n! i! (n-i)!}$$

$$= \sum_{i=0}^n \frac{1}{n!} \binom{n}{i}$$

$$= \frac{1}{n!} \sum_{i=0}^n \binom{n}{i}$$

$$= \frac{1}{n!} \left[\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} \right]$$

$$= \frac{2^n}{n!}$$

$$\text{(ii) } \sum_{i=0}^n \frac{(-1)^i}{i!(n-i)!} = \sum_{i=0}^n \frac{n! (-1)^i}{n! i! (n-i)!}$$

$$= \sum_{i=0}^n \frac{(-1)^i}{n!} \binom{n}{i}$$

$$= \frac{1}{n!} \sum_{i=0}^n (-1)^i \binom{n}{i}$$

$$= \frac{1}{n!} \left[\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} \right]$$

$$= 0.$$

2. S.T for all positive integers m and n , $n \binom{m+n}{m} = (m+1) \binom{m+n}{m+1}$

$$\text{Pf: } n \binom{m+n}{m} = n \frac{(m+n)!}{m! n!} = \frac{(m+n)!}{m! (n-1)!} = \frac{(m+1)(m+n)!}{(m+1)m! (n-1)!}$$

$$= (m+1) \frac{(m+n)!}{(m+1)! (n-1)!}$$

$$= (m+1) \binom{m+n}{m+1}.$$

3. Determine x if $\sum_{k=0}^{50} \binom{50}{k} 8^k = x^{100}$.

Sol:- we know that $\sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = (x+y)^n$.

let $x=8, y=1, n=50$

$$\begin{aligned} \therefore \sum_{k=0}^{50} \binom{50}{k} 8^k (1)^{50-k} &= (8+1)^{50} \\ &= (9)^{50} = \left[(\pm 3)^2 \right]^{50} \\ &= (\pm 3)^{100} \end{aligned}$$

Therefore $x = \pm 3$.

Fundamental Principles of Counting

The study of discrete and Combinatorial mathematics begins with two basic principles of Counting: the rules of Sum and product.



The Rule of Sum

If a first task can be performed in ' m ' ways, while a second task can be performed in ' n ' ways, and the two tasks cannot be performed at the same time, then there are ' $m+n$ ' ways to choose one of these tasks. In other words, if there are ' $n(A)$ ' ways to do ' A ' and distinct from them, ' $n(B)$ ' ways to do ' B ', then the number of ways to do A or B is $n(A) + n(B)$.

eg: A College library has 40 text-books on Computer Science and 50 textbooks dealing with Mathematics. Then the number of ways to select a book, which is either Computer Science or Mathematics is $50 + 40 = 90$.



Extension of Sum Rule

If tasks $T_1, T_2, \dots, T_m$ can be done in $n_1, n_2, \dots, n_m$ ways respectively and no two of these tasks can be performed at the same time, then the number of ways to do one of these tasks is $n_1 + n_2 + \dots + n_m$.

eg: If a student can choose a project either 20 from Mathematics or 35 from Computer Science or 15 from Engineering, then the student can choose a project in 70 ($20 + 35 + 15$) ways.



The Rule of Product

If a procedure can be broken down into first and second stages, and if there are ' m ' possible outcomes for the first stage and if, for each of these outcomes, there are ' n ' possible outcomes for the second stage, then the total procedure can be carried out, in the designated order, in ' mn ' ways. In other words, if there are $n(A)$ ways to do A and $n(B)$ ways to do B , then the number of ways to do A and B is $n(A) \cdot n(B)$.

eg: The drama club of Central University is holding tryouts for a Spring play. With '6' men and '8' women auditioning for the leading male and female roles, by the rule of product the director can cast his leading couple in $6 \times 8 = 48$ ways



Extension of Product Rule

Suppose a procedure consists of performing tasks $T_1, T_2, \dots, T_m$ in that order. Suppose task T_i can be performed in n_i ways after the tasks $T_1, T_2, \dots, T_{i-1}$ are performed, then the number of ways the procedure can be executed in the designated order is $n_1 n_2 n_3 \dots n_m$.

eg: "Estilocus" brand shirt available in 12 colors, has a male and female version. It comes in 4 sizes for each sex, comes in 3 makes of economy, standard and luxury. Then the number of different types of shirts produced are 288. ($12 \times 2 \times 4 \times 3$)

1. A tourist can travel from Hyderabad to Tirupati in 4 ways, (by plane, train, bus, taxi). He can travel then from Tirupati to Tirumala hills in 5 ways (by RTC bus, taxi, ropeway, motorcycle or walk). How many ways he can travel from Hyderabad to Tirumala hills?
2. Buick automobiles come in 4 models, 12 colors, 3 engine sizes, and 2 transmission types. (i) How many distinct Buicks can be manufactured? (ii) If one of the available colors is blue, how many different blue Buicks can be manufactured?
3. During a local campaign, 8 Republican and 5 Democratic candidates are nominated for president of the school board. (i) If the president is to be one of these candidates, how many possibilities are there for the eventual winner? (ii) How many possibilities exist for a pair of candidates (one from each party) to oppose each other for the eventual election?

4. A hotel offers 12 kinds of Sweets, 10 kinds of hot tiffins and 5 kinds of beverages (hot tea, hot Coffee, juice, Coke, ice cream). The breakfast consisting of a Sweet and a hot beverage or a hot tiffin and a Cold beverage. How many ways the breakfast can be ordered. (Ans: 54)

5. The board of directors of a pharmaceutical Corporation has 10 members. An upcoming stockholders meeting is scheduled to approve a new slate of Company officers (chosen from the 10 board members) (i) How many different slates consisting of a president, Vice president, Secretary and treasurer can the board present to the stockholders for their approval? (ii) Three members of the board of directors are physicians. How many slates from part (i) have (a) a physician nominated for the presidency? (b) At least one physician appearing on the slate?

Permutations

A permutation is an arrangement of objects in a definite order.



Permutation of 'n' different objects

The number of permutations of 'n' different objects taken all at a time, denoted by the symbol $nP_n = P(n, n) = n!$.

The number of permutation of 'n' different objects taken 'r' at a time, when $0 < r \leq n$ denoted by $nP_r = P(n, r) = \frac{n!}{(n-r)!}$.

The number of permutations of 'n' different objects taken all at a time, when repetition of objects allowed is n^n .

The number of permutations of 'n' different objects taken 'r' at a time, when repetition of objects allowed is n^r .

1. List all the permutations for the letters a, c, t
Sol: (i) a, c, t (ii) a, t, c (iii) c, t, a (iv) c, a, t (v) t, c, a (vi) t, a, c.

2. How many permutations are there for the Eight letters a, c, f, g, i, t, w, x? How many of these (i) start with the letter 't'.
(ii) start with the letter 't' and end with the letter 'c'.

Sol: we have the number of permutations of 8 letters taken all at a time = $8!$.

(i) Out of these $8!$ permutations, the number of permutation start with the letter 't' = $7!$ (t-----)

(ii) The number of permutations start with the letter 't' and end with the letter 'c' = $6!$

3. How many 3-digit numbers can be formed from the digits 1, 2, 3, 4 & 5 assuming that (i) repetition of digits allowed (ii) repetition of digits not allowed.
(Ans: (i) 125 (ii) 60)

4. How many 4-digit numbers can be formed from the digits 1, 2, 3, 4 using the digit without repetition?
(Ans: 24)

5. How many 5-digit telephone numbers can be constructed using the digits 0 to 9 if each number starts with '67' and no digit appears more than once? (Ans: 336)
6. How many 3-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5 and 6 if each digit can be used only once? (Ans: 180)
7. How many 3-digit even numbers can be made using the digits 1, 2, 3, 4, 6, 7 if no digit is repeated? (Ans: 60)
8. How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5, 6 if the digits can be repeated. (Ans: 108)
9. How many 4-digit numbers that can be formed using the digits 1, 2, 3, 4, 5 if no digit is repeated. How many of them will be even (Ans: 120, 48)
10. How many 5-digit numbers greater than 30000 can be formed from the digits 1, 2, 3, 4, 5? (Ans: 72)
11. How many 4-digit numbers divisible by 4 can be made with the digits 1, 2, 3, 4, 5 if the repetition of digits is not allowed? (Ans: 24)
12. How many 3-digit numbers can be formed from the digits 1, 2, 3, 4, 5, using the digits without repetition. (i) How many of them are even (ii) How many are even and greater than 300 (Ans: 60, 24, 15)
13. How many words, with or without meaning can be made from the letters of word MONDAY, assuming that no letter is repeated if (i) 4 letters are used at a time (ii) all letters are used at a time (iii) all letters are used but first letter is a vowel (Ans: 360, 720, 240)
14. In how many ways can the symbols a, b, c, d, e, e, e, e be arranged so that no 'e' is adjacent to another 'e'. (Ans: 24)

15. Find the value of 'n' in each of the following

(i) $P(n, 2) = 90$ (ii) $P(n, 3) = 3 P(n, 2)$ (iii) $2 P(n, 2) + 50 = P(2n, 2)$

(iv) $(n-1)P_3 : nP_4 = 1 : 9$

(Ans (i) $n=10$ (ii) $n=5$ (iii) $n=5$
(iv) $n=9$)

16. Find 'r' if (i) $5P_r = 2 \cdot 6P_{r-1}$ (ii) $5P_r = 6P_{r-1}$ (Ans (i) $r=3$
(ii) $r=4$)

Permutation when all the objects are not distinct objects

The number of permutations of 'n' objects where 'p' objects are of the same kind and rest are all different = $\frac{n!}{p!}$

The number of permutations of 'n' objects where 'p₁' object are of the one kind, 'p₂' are of second kind, ..., 'p_k' are of kth kind and the rest, if any, are of different kind = $\frac{n!}{p_1! \cdot p_2! \cdot \dots \cdot p_k!}$

1. Find the number of permutations of the letters of the word 'ALLAHABAD'.

Sol: Here there are 9 objects (letters) of which there are four A's and two L's and rest are all different.

Therefore the number of permutations = $\frac{9!}{4! \cdot 2!} = 7560$.

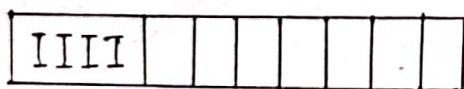
2. Find the number of (linear) arrangements of the four letters in 'BALL' (Ans: 12)

3. How many permutations of the letters in 'MISSISSIPPI'. In how many of these arrangements have the four I's not come together.

Sol: Here there are 11 objects (letters) of which there are four I's, four S's and two P's

Therefore the number of permutations = $\frac{11!}{4! \cdot 4! \cdot 2!} = 34650$.

The number of arrangements with the four I's come together = $\frac{8!}{4! \cdot 2!} = 840$.



∴ the number of permutation with the four I's not come together = $34650 - 840 = 33810$.

4. Find the number of linear arrangements of the nine letters in DATABASES (Ans: 30240)

5. Find the number of arrangements of the letters in TALLAHASSEE (Ans: 821600).

6. How many ways can the letters in DATAGRAM be arranged. For these arrangements, how many have all three A's together

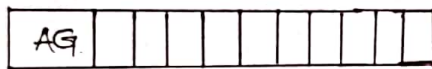
(Ans: 6720, 720)

7. Find the number of linear arrangement of all letters in the word MASSASAUGA. Also find the number of linear arrangements in which all four A's are together. (Ans: 25200, 840)
8. How many ways can the letters of the word ASSASSINATION be arranged so that all the S's are together. (Ans: 151200)
9. How many arrangements are there of all the letters in SOCIOLOGICAL.
 (i) In how many of the arrangements are A and G adjacent. (ii) In how many of the arrangements are all the Vowels adjacent.

Sol: Here there are 12 objects (letters) of which there are three O's, two C's, two I's and two L's and the rest are all different.

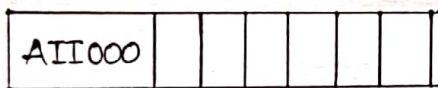
$$\text{Therefore the number of arrangements (permutations)} = \frac{12!}{3! \cdot 2! \cdot 2! \cdot 2!} = 9979200.$$

$$\text{The number of arrangements with A \& G adjacent} = \frac{11!}{3! \cdot 2! \cdot 2! \cdot 2!} \times 2!$$



$$= 1663200.$$

$$\text{The number of arrangements with Vowels adjacent} = \frac{7!}{2! \cdot 2!} \times \frac{6!}{3! \times 2!}$$



$$= 75600.$$

10. Find the number of arrangements of the letters of the word INDEPENDENCE. In how many of these arrangements
 (i) do the words start with 'P' (ii) do all the vowels always occur together (iii) do the vowels never occur together (iv) do the words begin with 'I' and end in 'P'.
 (Ans: 1663200, 138600, 16800, 1646400, 12600)

11. If the different permutations of all the letters of the word EXAMINATION are listed as in a dictionary, how many words are there in list before the first word starting with E. (Ans: 907200)

12. In how many ways can the letters of the word PERMUTATIONS be arranged if the (i) words start with 'P' and end with 'S'.
 (ii) Vowels are all together (ii) there are always 4 letters P and 'S'.
 (Ans: 1814400; 2419200; 25401600)

13. Determine the number of (Staircase) paths in the xy-plane from (2,1) to (7,4), where each such path is made up of individual steps going one unit to the right (R) or one unit upward (U).

Sol:

The number of paths emerges as the number of arrangements of the five R's and three U's, which is $\frac{8!}{5!3!} = 56$.

14. How many three digit numbers can be made from the digits 1, 3, 3, 4.

Sol: Here 134 (or) 133 (or) 433

$$\text{ie } 3! + \frac{3!}{2!} + \frac{3!}{2!} = 6 + 3 + 3 = 12$$

That are 134, 143, 341, 314, 413, 431, }
133, 313, 331 } = 12.
433, 343, 334 }

15. How many 4 digit numbers can be made from the digits 1, 2, 3, 7, 7 and 8.

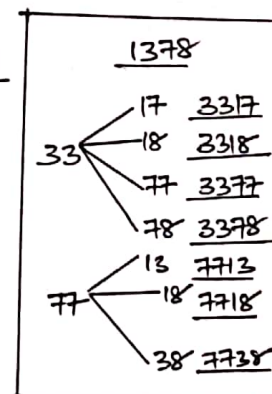
Sol: Here 1378 (or) 3317 (or) 3318 (or) 3377 (or) 3378 (or)
7713 (or) 7718 (or) 7738

$$= 4! + \frac{4!}{2!} + \frac{4!}{2!} + \frac{4!}{2!2!} + \frac{4!}{2!} + \frac{4!}{2!} + \frac{4!}{2!} + \frac{4!}{2!}$$

$$= 4! + 6 \left(\frac{4!}{2!} \right) + \frac{4!}{2!2!}$$

$$= 24 + 72 + 6$$

$$= 102.$$



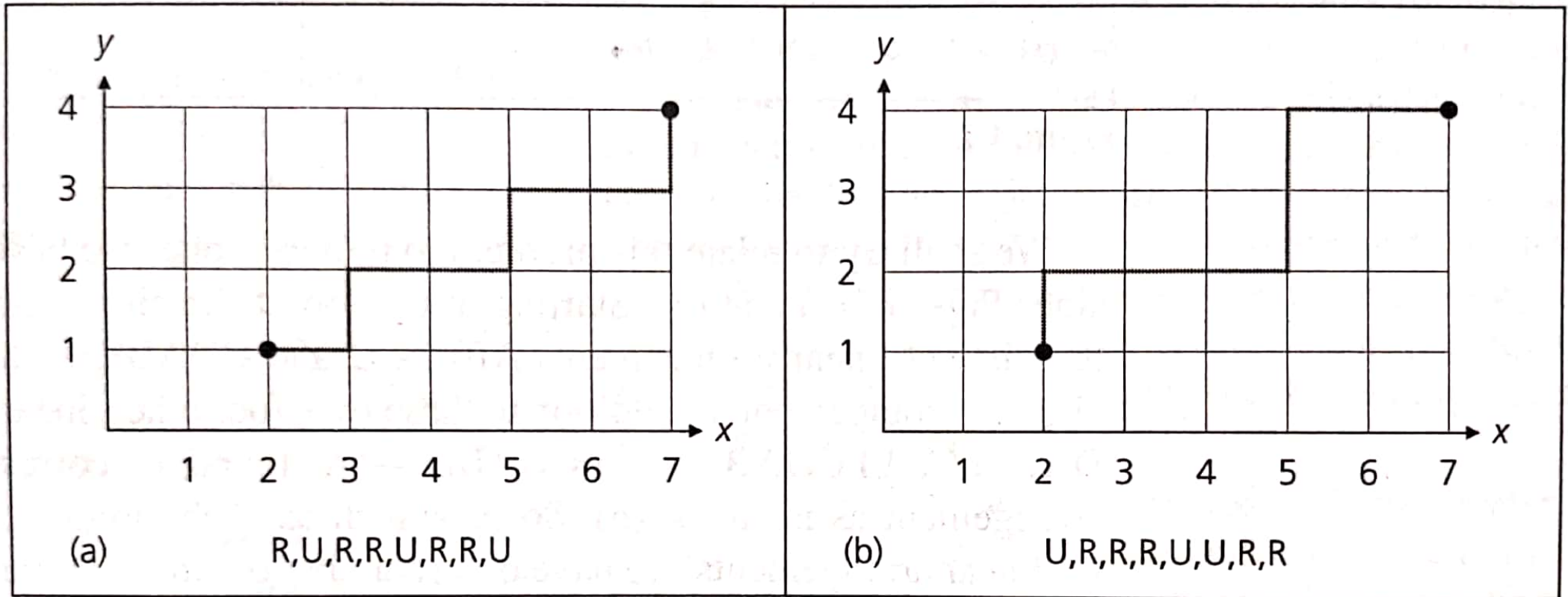


Figure 1.1